Ufuk Usubutun

New York, NY usubutun@nyu.edu (646) 217-9430 ufukusubutun.github.io

Summary

Ph.D. candidate and network systems engineer with hands-on experience in **cross-layer protocol evaluation and system modeling**. Strong background in **network testbed experimentation, queueing theory, Markov modeling,** and **transport protocols**. Experienced in interpreting **standards** across IETF (wide-area), 3GPP (cellular), and Ultra Ethernet (data-center) ecosystems, and collaborating with AT&T Labs Research and Nokia Bell Labs. (F1 Visa Holder.)

Education

New York University, PhD in Electrical Engineering — Brooklyn, NY — Sept 2020 – May 2026 (Expected)

Supervisor: Prof. Shivendra Panwar

Bilkent University, BSc in Electrical and Electronics Engineering — Ankara, Turkey — Sept 2015 – Jan 2020

Minor in Political Science, Exchange Semester in AGH University, Krakow, Poland — GPA: 3.76/4.0

Experience

New York University, Research Assistant / Fellow — Brooklyn, NY — Sept 2020 – ongoing

- Evaluated TCP resilience to packet reordering through testbed experiments and queuing theory; demonstrated feasibility of per-packet load-balancing over wireless multipath, and of simpler switch designs through relaxation of in-sequence delivery.
- Collaborated with AT&T Labs Research to model reliability bottlenecks of disaggregated RANs, including wireless backhaul, and to optimize activation of 5G dual-connectivity.
- Assisted instruction in graduate courses on Computer Networks and Network Modeling.

Marvell Semiconductor, Switch Architect Engineer Intern — Santa Clara, CA — Jun 2024 – Aug 2024

- Interpreted Ultra Ethernet Transport (UET) standard drafts and delivered technical briefings to the data center switch architecture team.
- Evaluated proposed UET congestion-control protocols through simulations.
- Built Markov models to guide buffer sizing decisions in the Teralynx switch architecture.

AT&T Labs Research, Summer Research Intern — Bedminster, NJ — Jun 2023 - Aug 2023

- Modeled inactivity timer-based 5G RRC state transitions to improve energy efficiency.
- Filed patent covering potential implementations with Open-RAN controllers.

Nokia Bell Labs, Networking Research Intern — Murray Hill, NJ — Jun 2022 - Aug 2022

- Formulated a segment routing (SRv6/MPLS) compatible, traffic oblivious routing problem.
- Implemented a scalable, descent-based algorithm with *pytorch* to solve the problem.

Darkblue Telecommunication Systems, Project Engineer — Ankara, Turkey — Feb 2020 - Aug 2020

- Implemented MATLAB workflow for positioning UAVs using LTE Channel Reference Signals.
- Collected field samples using software-defined radios.

Fraunhofer Institute for Integrated Circuits, Undergraduate Research Intern — Erlangen, Germany — Jun 2019 - Sept 2019

- Developed an OFDM-based cooperative communications simulator in Python.

Technical Skills

Expertise Areas: TCP congestion control and loss detection, queueing theory, standards (IETF RFCs, 3GPP, Ultra Ethernet), Markov modeling, network experiment design.

Programming Experience: Python (pandas, simulation, optimization), Bash scripting for testbeds, MATLAB (Markov modeling), familiarity with Linux TCP internals (net/ipv4), measurement and experimentation with tc, iptables, tshark.

Coursework: Network Modeling, Data-Center Networking, Network Optimization, Machine & Reinforcement Learning.

Publications

Designing Ultra-Reliable High-Capacity Wireless Backhaul – with AT&T — IEEE WCL (in review)
*Ufuk Usubutun**, Andre Gomes*, Shankar Narayanan, Matti Hiltunen, Shivendra Panwar

Backbone Switches No Longer Need to Deliver Packets in Sequence — IEEE OJCOMS (in review)
Ufuk Usubutun, Fraida Fund, Shivendra Panwar

Modeling and Optimizing Dual-Connectivity Activation in Cellular Networks – with AT&T — (in revision)
Caglar Tunc, *Ufuk Usubutun*, Yuxuan Jiang, Shivendra Panwar

Designing Reliable Virtualized Radio Access Networks – with AT&T — IEEE Globecom 2024
Ufuk Usubutun, Andre Gomes, Shankar P. Narayanan, Matti Hiltunen, Shivendra Panwar

Oblivious Routing Using Learning Methods – with Nokia Bell Labs — IEEE Globecom 2023
Ufuk Usubutun, Murali Kodialam, T.V. Lakshman, Shivendra Panwar

Do Switches Still Need to Deliver Packets in Sequence? — IEEE HPSR 2023
Ufuk Usubutun, Fraida Fund, Shivendra Panwar — Best Paper Award

Patents

Methods, Systems, and Devices For Determining Inactivity Timer Values in Mobile Networks — Filed Oct 2024
Ufuk Usubutun, Yuxuan Jiang, Caglar Tunc, Xuan Tuyen Tran, Aleksandr Zelezniak, Yu Zhou

Awards & Recognitions

Dante Youla Award for Graduate Research Excellence – NYU ECE Department — 2024
Best Paper Award – IEEE Conference on High Performance Switching and Routing — 2023
Oustanding Innovation Award – Nokia Global Student Program — 2022

Languages

English (Fluent), French (Proficient, DELF B2), Turkish (Native)

Leadership & Service

- Mentor, NYU ARISE High School Research Program – guided student projects involving network testbed experiments.
- Mentor, FRC Robotics Team 3390 Anatolian Eaglebots – taught Java programming to high school students.
- Co-lead, Bilkent Sociology Club - organized seminars, discussion sessions and city walks guided by academics.
- Member, Bilkent Improvisation Club - directed improvisation sessions and performed in them.

Last updated in March 2026